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Security Operations Center (SOC) Investigation Report

-Investigated by
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Investigation Title

Windows Active Directory Traffic Investigation

1. Overview

In this investigation, I analyzed a PCAP file using Wireshark with the goal of understanding host communications, protocol usage, and identifying whether the observed activity was normal network behavior or potentially malicious.

Unlike previous investigations, the focus was on understanding the environment and protocol interactions rather than immediately searching for indicators of compromise.

2. Capture Statistics

Metric	Value
Total Packets	51,181
Capture Size	27 MB
Average Packet Size 516 Bytes	
Duration	10 Minutes 23 Seconds
Start Time	2026-01-28 04:34:03
End Time	2026-01-28 04:44:27

3. Initial Observations

The majority of communication involved two internal hosts:

- 10.1.21.58
- 10.1.21.2

External communication was observed with Cloudflare, Google, Hostinger, Akamai, and Microsoft infrastructure.

Protocol hierarchy analysis showed significant use of:

- SMB2
- DCE/RPC
- LSARPC
- SAMR
- Kerberos
- LDAP
- TLS

The protocol distribution suggested Windows domain-related activity rather than normal web browsing traffic.

4. Host Communication Analysis

The most active conversation was:

10.1.21.58 ↔ 10.1.21.2

Analysis of TCP conversations revealed communication over:

Port Protocol

445 SMB

389 LDAP

135 RPC Endpoint Mapper

88 Kerberos

These services are commonly associated with Active Directory environments.

SMB Investigation

SMB2 analysis revealed:

- Tree Connect Requests
- IPC\$ share access
- SAMR activity
- User and domain enumeration operations

Observed operations included:

- EnumDomains
- LookupDomain
- OpenDomain
- LookupNames
- OpenUser
- QueryUserInfo
- GetGroupsForUser

A Tree Connect request to:

`\WIN-LU4L24X3UB7.win11office.com\IPC$`

was identified.

This indicated communication with a Windows host within the domain environment.

Kerberos Investigation

Kerberos traffic showed:

- AS-REQ
- AS-REP
- TGS-REQ
- TGS-REP

These are standard authentication exchanges used by Active Directory.

A Kerberos response containing:

KRB5KDC_ERR_PREAUTH_REQUIRED

was observed.

Further research showed this is commonly seen during normal Kerberos authentication workflows and is not necessarily indicative of malicious activity.

LDAP Investigation

LDAP traffic included:

- bindRequest
- bindResponse
- searchRequest

The communication pattern suggested directory service queries between systems.

Although several packets appeared as malformed in Wireshark, the overall behavior remained consistent with Active Directory-related operations.

Findings

Confirmed

- Active communication between two internal systems.
- SMB, LDAP, Kerberos, and RPC services were actively used.
- Domain-related account and group queries were observed.
- Windows administrative shares (IPC\$) were accessed.
- Kerberos authentication exchanges occurred successfully.

Not Confirmed

- Malware activity.
- Data exfiltration.
- Command-and-control communication.
- Lateral movement.
- Malicious file transfers.

Key Learning Outcomes

This investigation highlighted the importance of understanding Windows networking protocols before making security conclusions.

Major concepts encountered:

- SMB
- LDAP
- Kerberos
- Active Directory
- RPC
- SAMR

A key lesson learned was that seeing account enumeration activity does not automatically indicate malicious behavior. Similar activity can be generated by normal Windows domain operations.

Conclusion

Based on the evidence collected, the traffic appears consistent with a Windows host interacting with Active Directory services within a domain environment.

No clear indicators of compromise were identified during the investigation. The exercise served primarily as an opportunity to understand how SMB, LDAP, Kerberos, and Active Directory-related protocols appear within packet captures and how they interact during normal Windows operations.